

TECH OF



ANÁLISE TÁXIS NOVA YORK

RITA LIMA



TECH OF

CONTEXTO NEGÓCIO

- O serviço é baseado em motoristas independentes e vendedores que geram receita compartilhada por corrida.
- Alta densidade populacional e fluxo turístico tornam o mercado de táxis em NYC altamente competitivo e eficiente.
- A análise de dados ajuda a entender padrões de procura, comportamento de clientes e rentabilidade no negócio.
- Vendor foca em volume e distribuição; motoristas buscam rentabilidade e eficiência operacional.



NYC TAXI DATA ANALYSIS



TRANSFORMAÇÃO DADOS BRUTOS

O projeto demonstra a transformação eficaz de dados brutos em informações acionáveis para suporte à decisão.



ESTRUTURA PROFISSIONAL DE BI

Análise estruturada com boas práticas de business intelligence, desde a preparação até à apresentação dos insights.



ESTUDO APLICADO REALISTA

Simulação de cenário empresarial real com múltiplos stakeholders para otimizar resultados com dados.



COMPETÊNCIAS TÉCNICAS E ANALÍTICAS

Projeto reflete habilidades em Power BI, pensamento crítico e comunicação orientada para negócios.



PREPARAÇÃO E MODELAÇÃO DOS DADOS

Limpeza e Análise Exploratória

Importação e análise exploratória dos dados para identificar inconsistências e garantir qualidade.

Criação de Colunas Derivadas

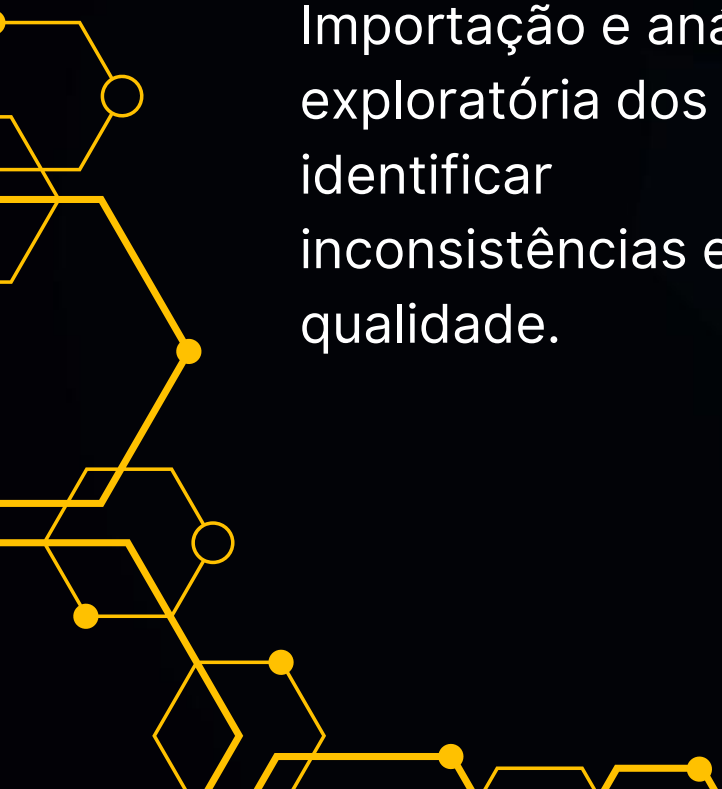
Novas colunas como hora de pickup e dia da semana foram criadas para análises temporais detalhadas.

Classificação e Transformação de Dados

Tipos de veículos classificados conforme passageiros, convertendo dados técnicos em categorias de negócio.

Modelação e Relacionamentos

Criação de relacionamentos entre tabelas seguindo modelo estrela para garantir cálculos corretos.





TAXI

VENDOR DASHBOARD

1.18

Average Passengers

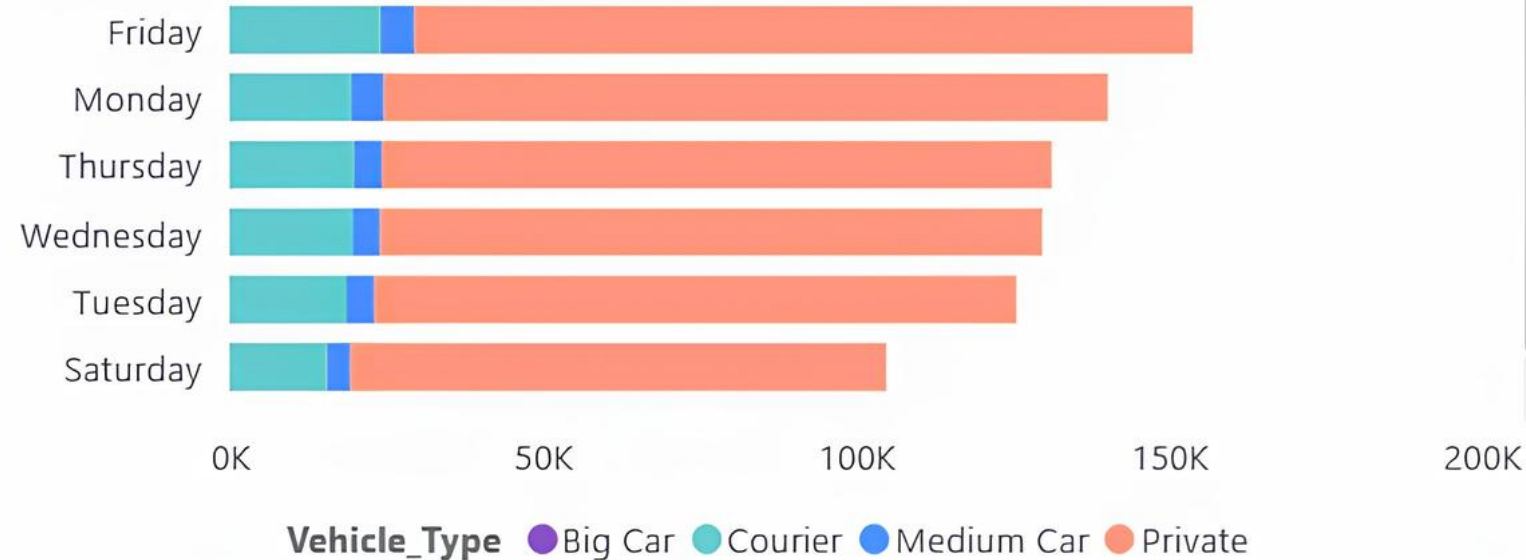
\$1.63

Average Tip

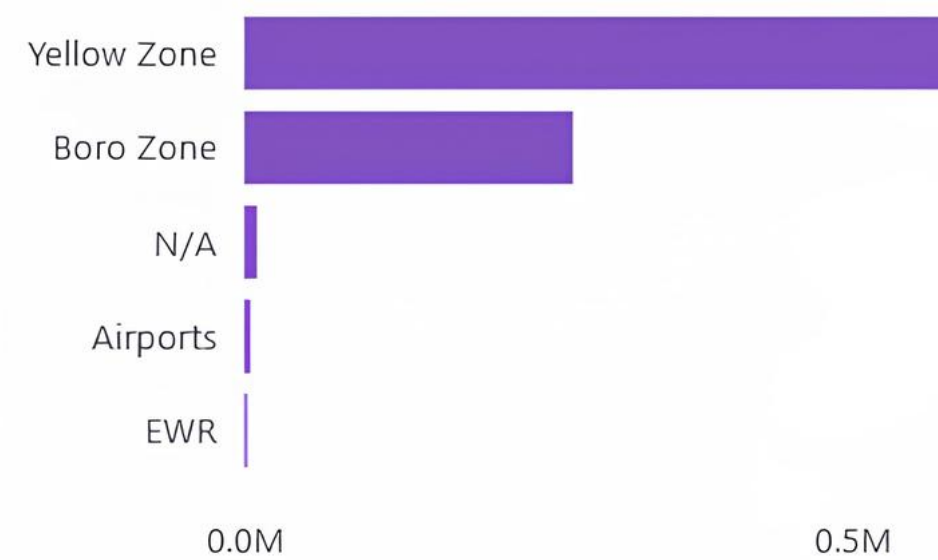
\$16.7M

Total Amount

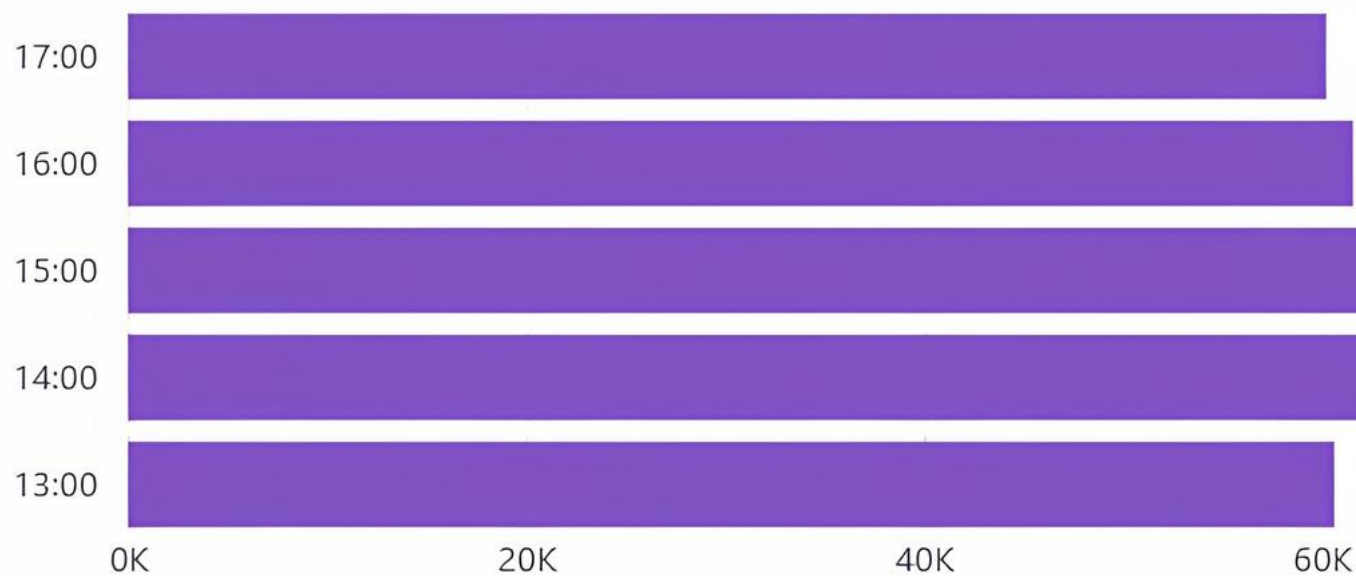
Vehicle Type By Day of the Week



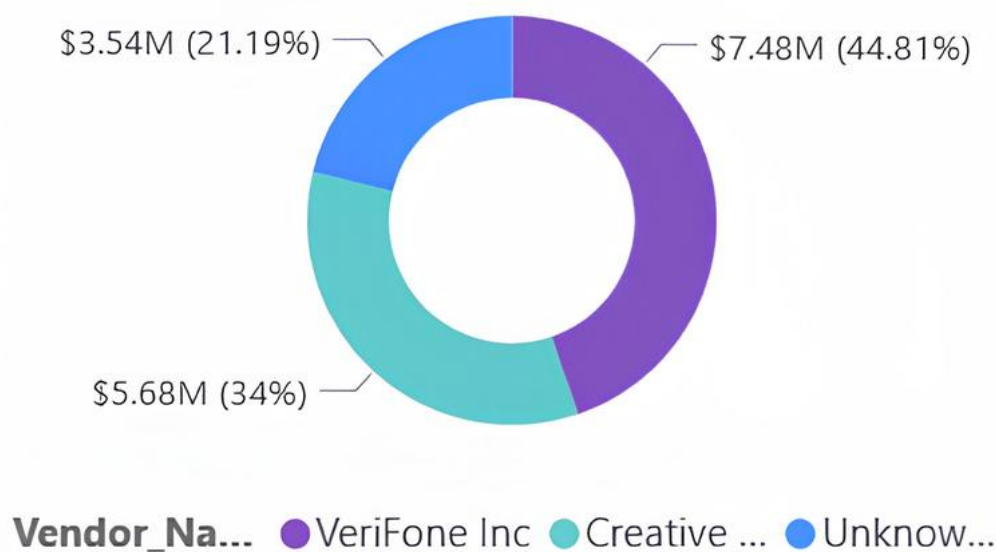
Nr. Trips Per Service Zone



Average Nr. Pickups Per Hour



Total Amount by Vendor



Vendor

All

Day of Week

All

Pickup Day

01/05/2020

30/06/2020

Vehicle Type

Big Car

Courier

Medium Car

Private

Payment Type

Cash

Credit Card

Dispute

Average Congestion Subcharge By Service Zone

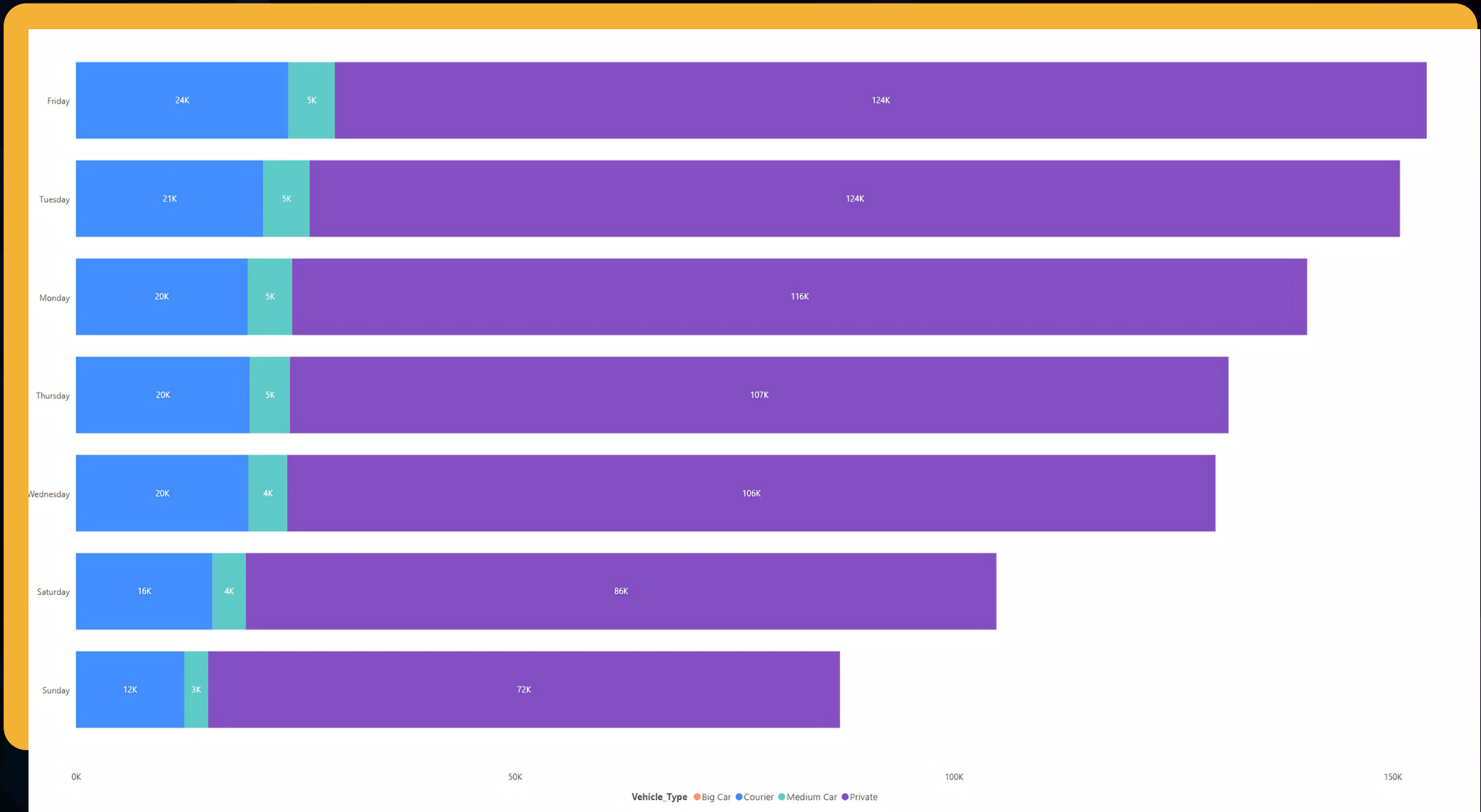




VENDOR DASHBOARD

Focus the fleet on private vehicles and staff up on weekdays, where over 80% of demand sits

WHAT IS THE DISTRIBUTION OF VEHICLE TYPES BY DAY OF THE WEEK?

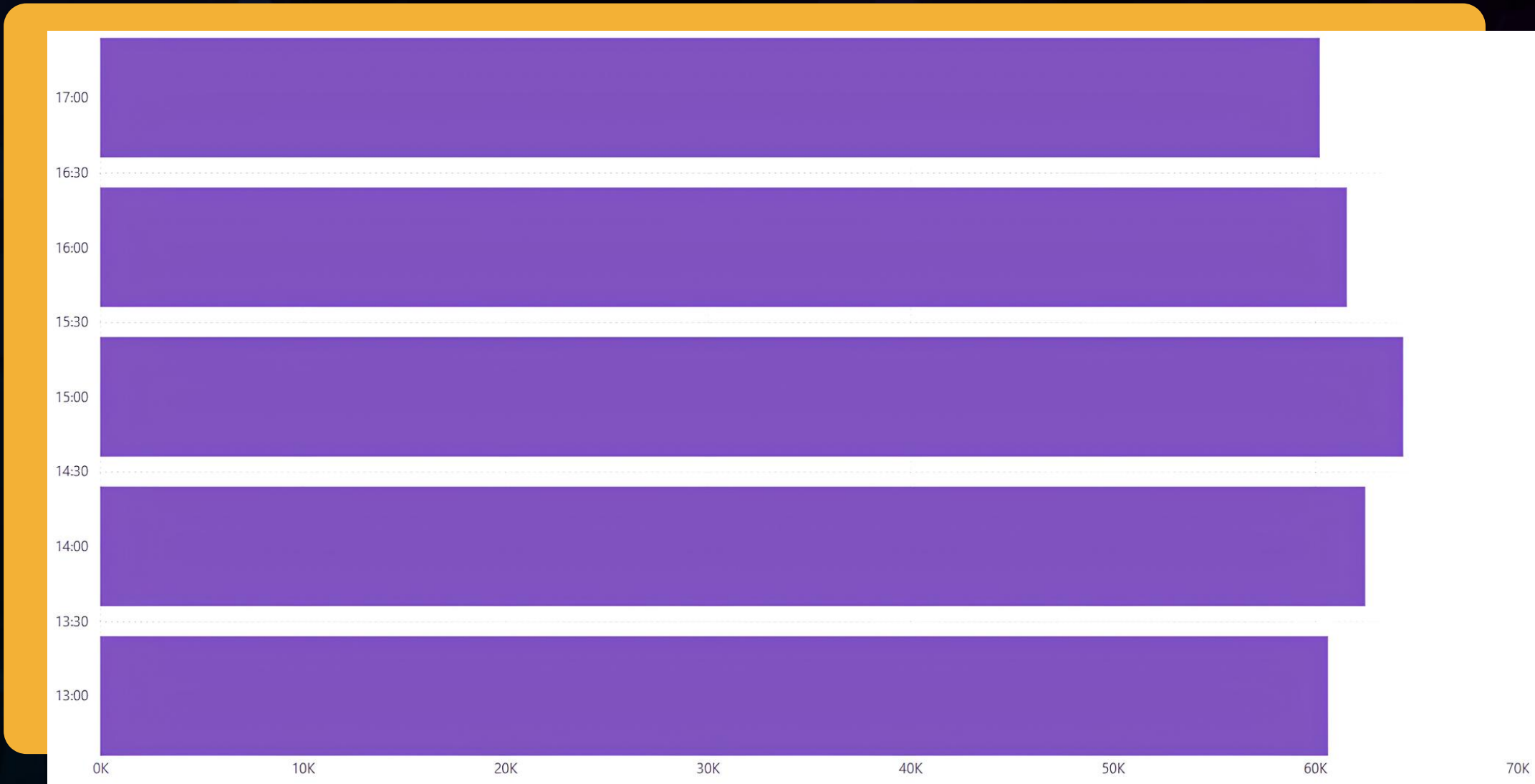




VENDOR DASHBOARD

Demand peaks across a sustained afternoon plateau (13:00–18:00, peaking at 15:00) — schedule shifts to maximise active drivers throughout this window and avoid mid-afternoon shift changes.

WHICH FIVE HOURS A DAY
HAVE THE MOST STARTING TIMES FOR RIDES?





WHICH SERVICE ZONES
GENERATE THE HIGHEST AVERAGE CONGESTION
SUBCHARGE

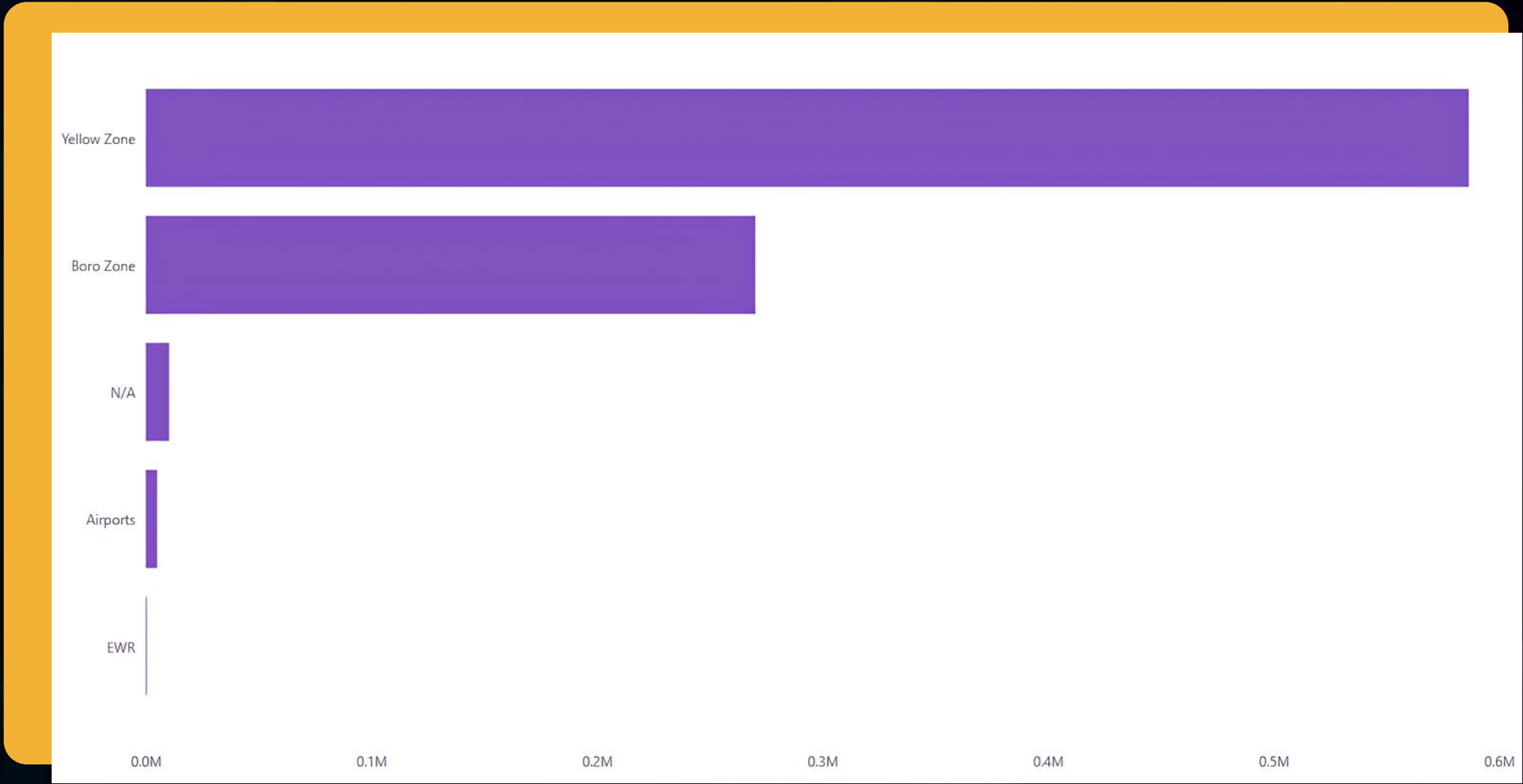
VENDOR
DASHBOARD

The Yellow Zone (Manhattan) carries by far the highest average congestion surcharge (~\$2.37), confirming central Manhattan as the operational core — but also flagging fare transparency as a customer-experience priority there.





WHICH SERVICE ZONES
GENERATE THE HIGHEST NUMBER OF TRIPS?



VENDOR
DASHBOARD

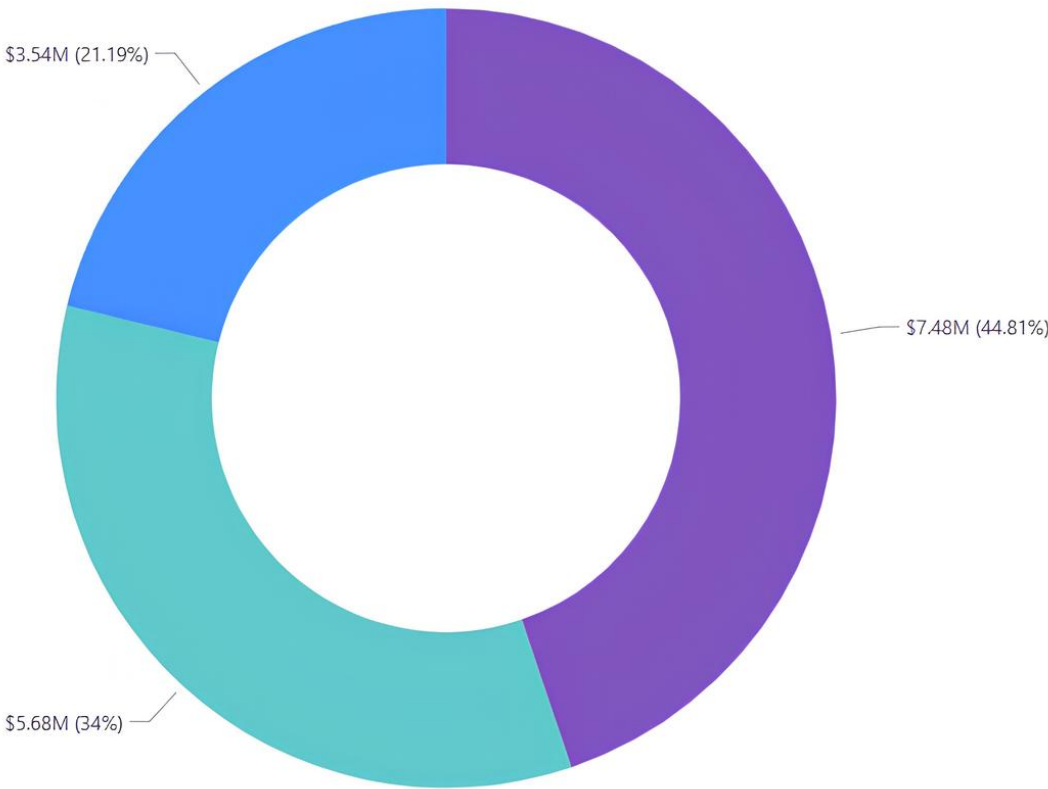
The Yellow Zone (Manhattan) and Boro Zone together account for ~98% of all trips (67% and 31%), confirming Manhattan as the operational core — anchor fleet supply there, while treating the low-volume, high-value airport segment as a targeted opportunity.



WHICH VENDOR
GENERATES THE HIGHEST TOTAL REVENUE?

VENDOR
DASHBOARD

VeriFone Inc is the top revenue vendor (~\$7.73M, 45%), but the 'Unknown Vendor' share (~\$3.62M, 21%) is a material data-governance gap that should be traced and resolved before revenue is treated as reliable.



Vendor_Name VeriFone Inc Creative Mobile Technologies, LLC Unknown Vendor



TAXI

EMPLOYEE DASHBOARD

\$19.13

Average Income

16

Average Nr Trips/day

5.82 Miles

Average Distance by Trip

Vendor

All

Pickup day

01/05/2020

30/06/2020

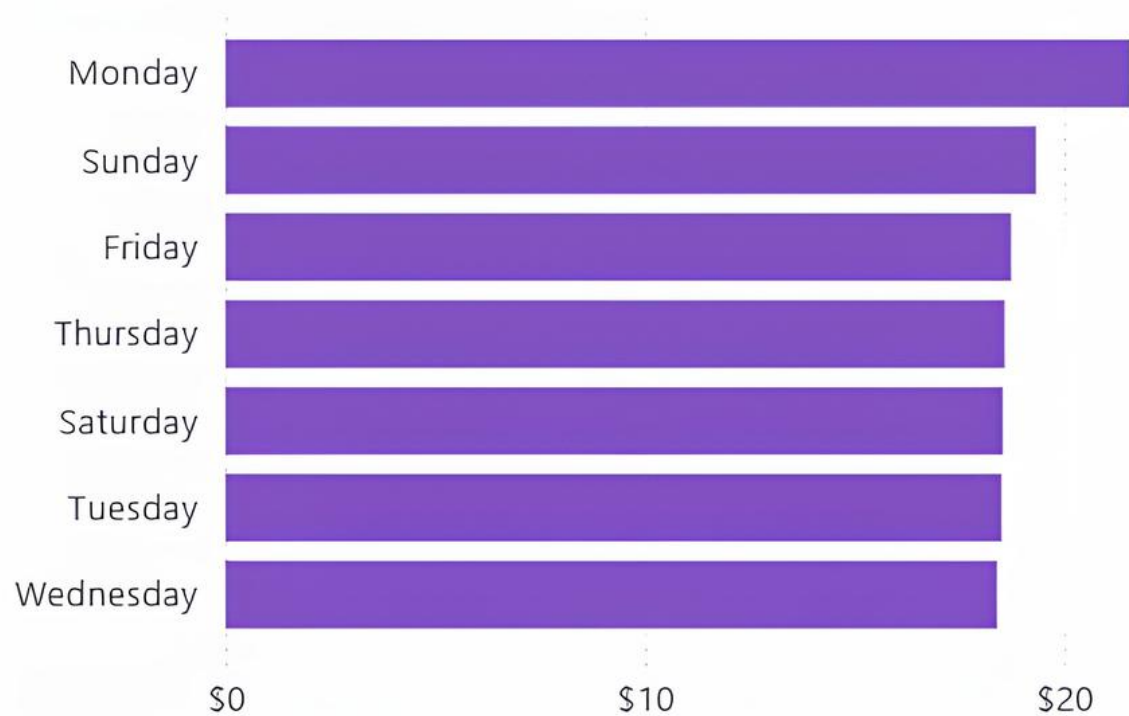
Day of week

All

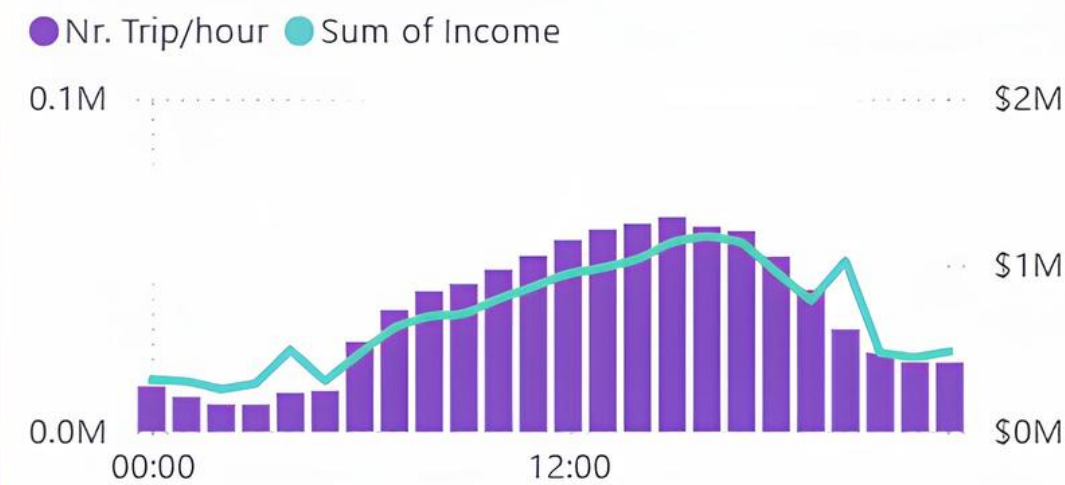
Pickup Time

All

Average Income/Day of Week



Nr. of Trips/Hour vs Sum of Income (May-June)



Big Car

Courier

Medium Car

Private

Cash

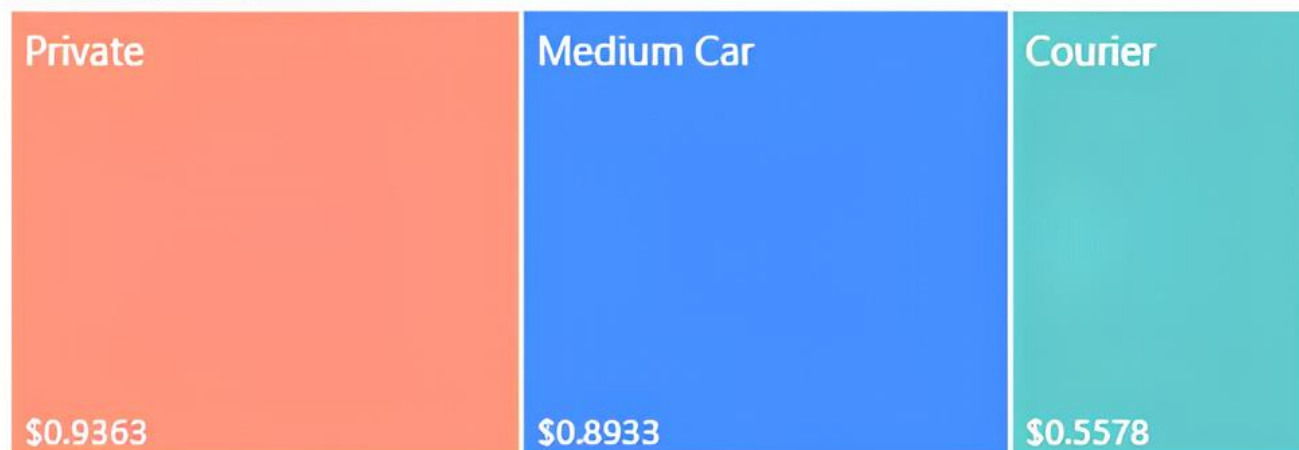
Credit Card

Dispute

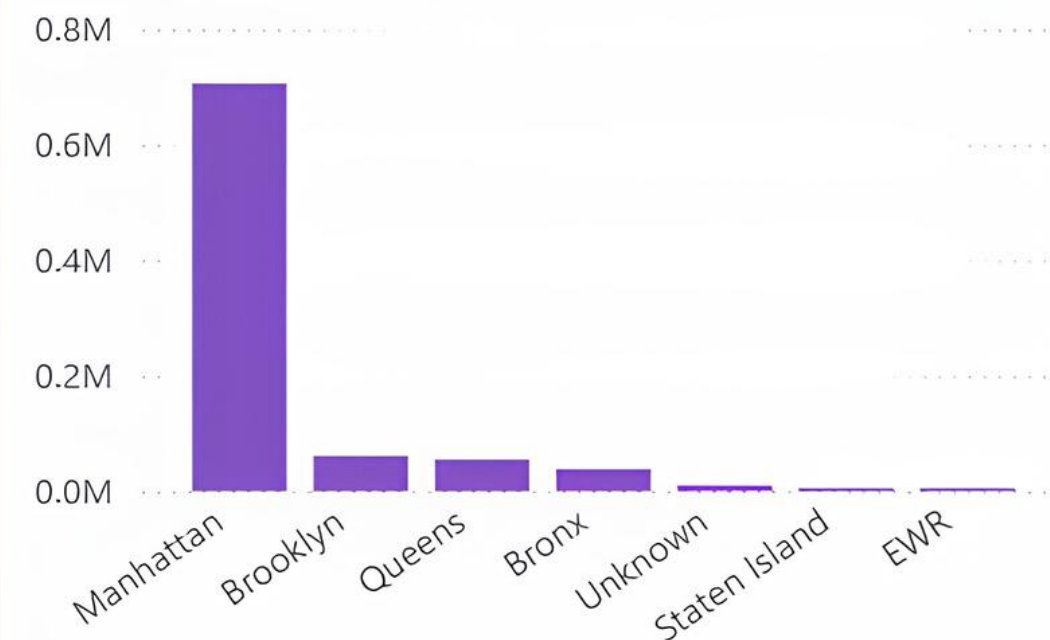
No Charge

Unknown

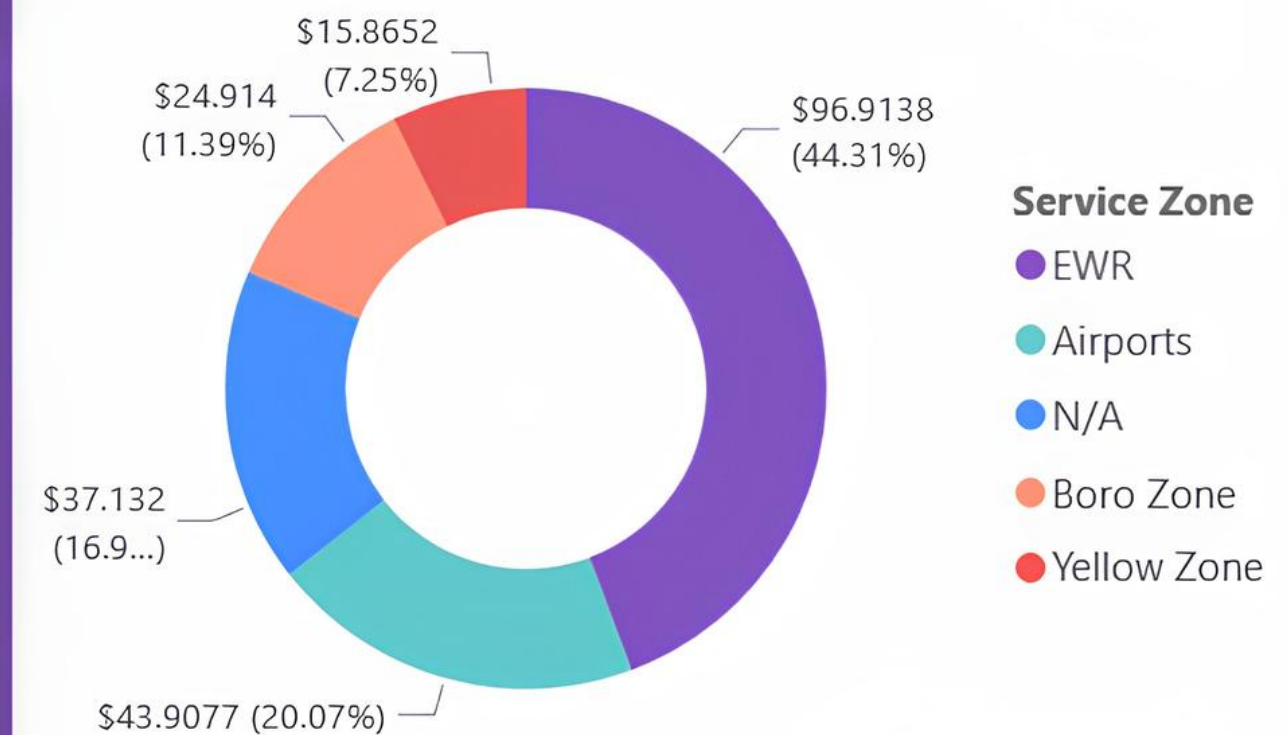
Average Tip/Mile



Nr. of Trips/Borough



Average Income/Zone

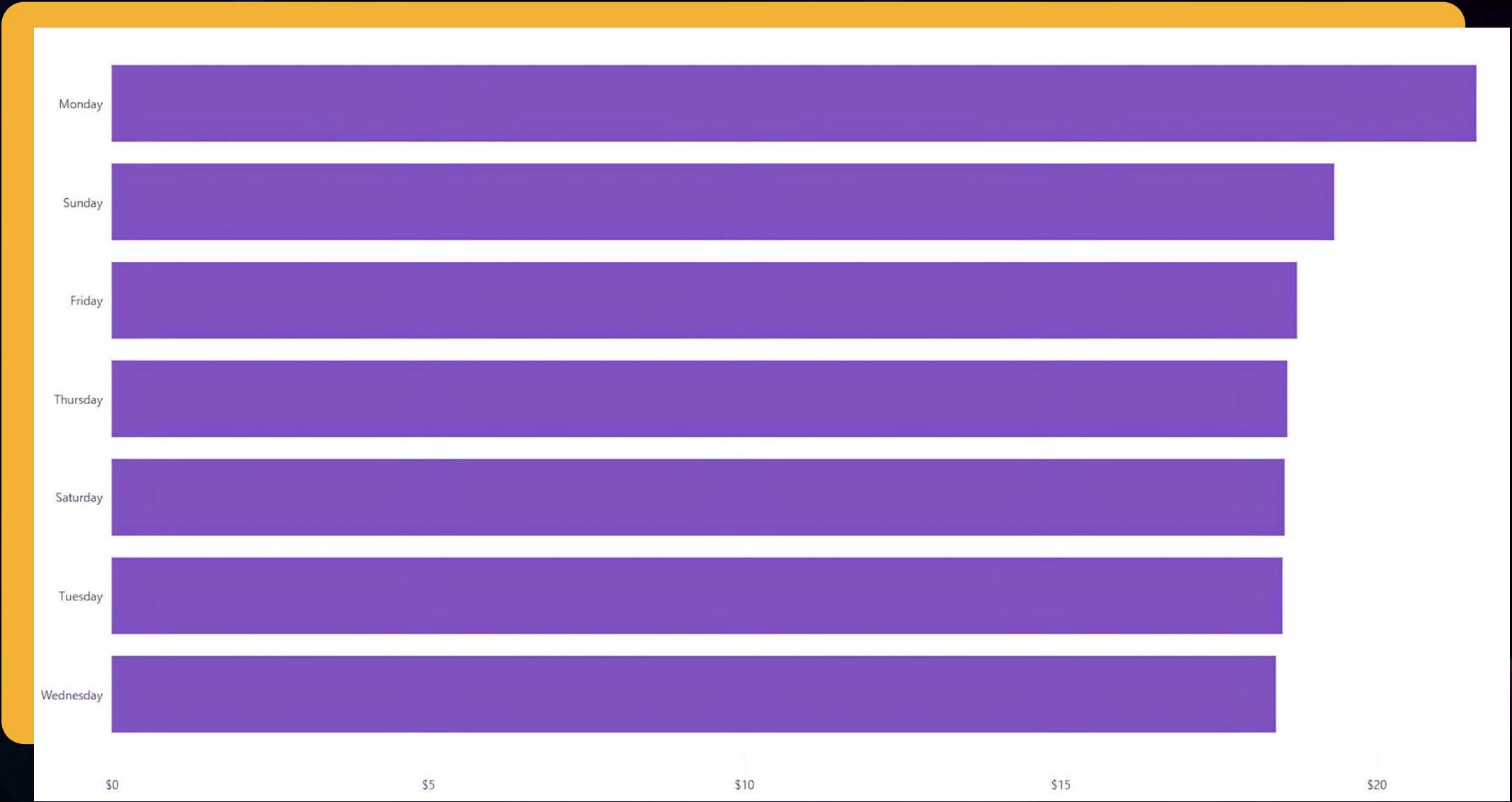




EMPLOYEE DASHBOARD

Prioritize working more hours on Mondays and Sundays to maximize productivity and earnings.

WHAT ARE THE TWO DAYS A WEEK WHEN
THE AVERAGE INCOME PER TRIP IS THE HIGHEST?

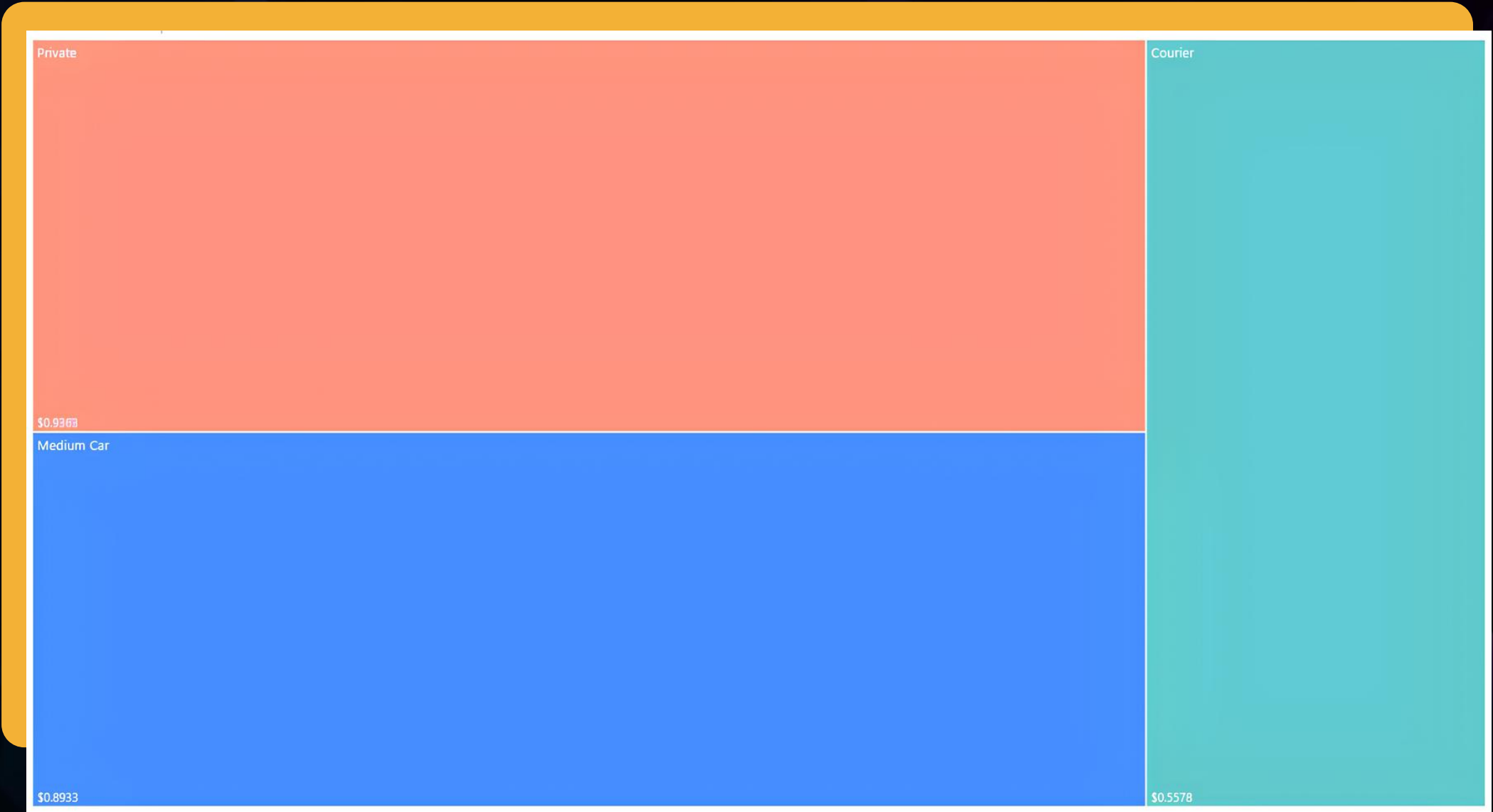




WHAT IS THE TYPE
OF VEHICLE WITH THE HIGHEST TIP PER MILE

EMPLOYEE DASHBOARD

Maximize earnings by completing more trips with a 5-seat vehicle, or by accepting longer trips with larger vehicles to increase tip income.

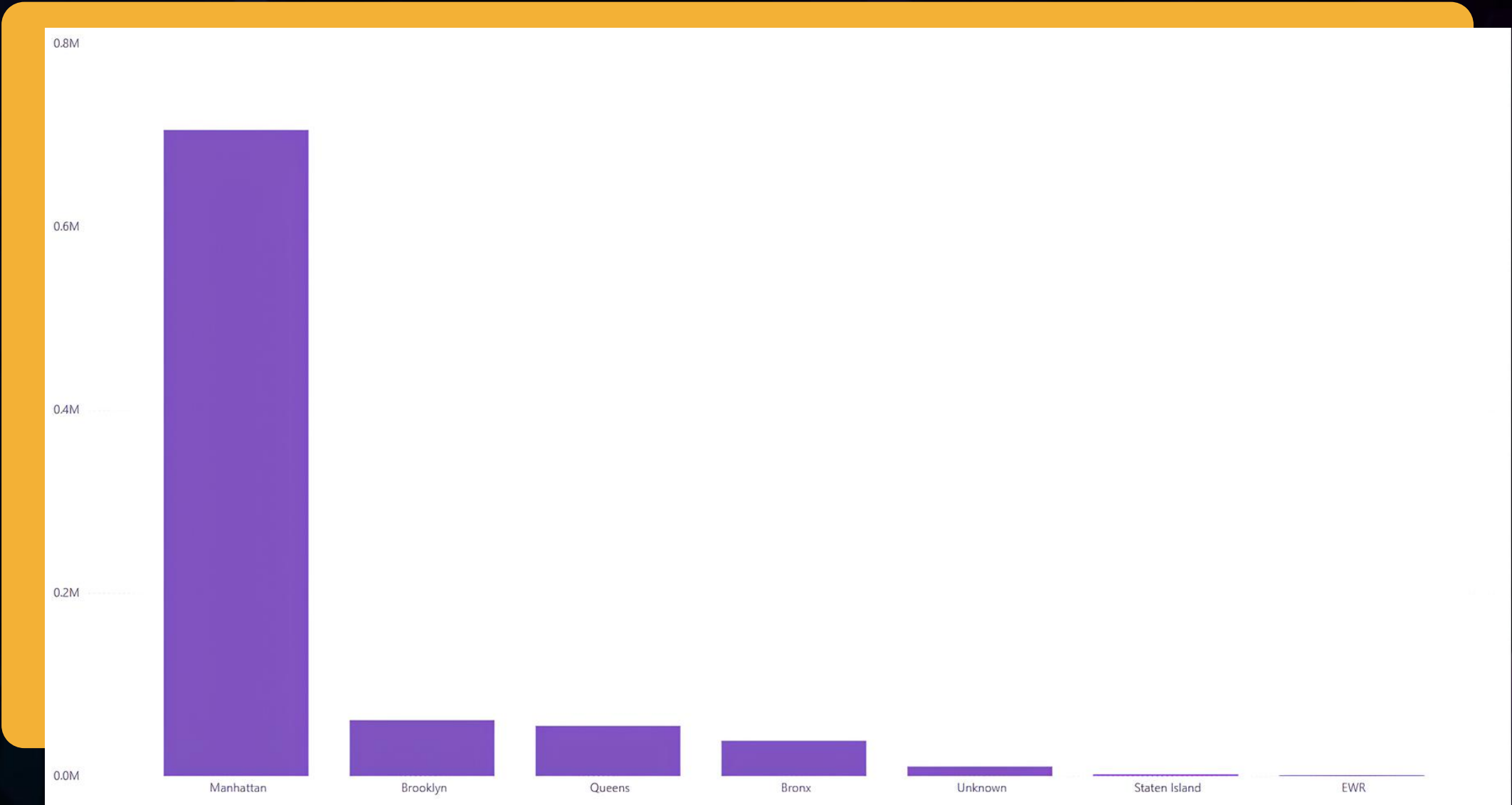




WHICH BOROUGH ZONES
RECORD THE HIGHEST NUMBER OF PICKUPS?

EMPLOYEE DASHBOARD

Focus on Manhattan, as it has the highest number of pickups and offers the best opportunity to maximize the number of trips.

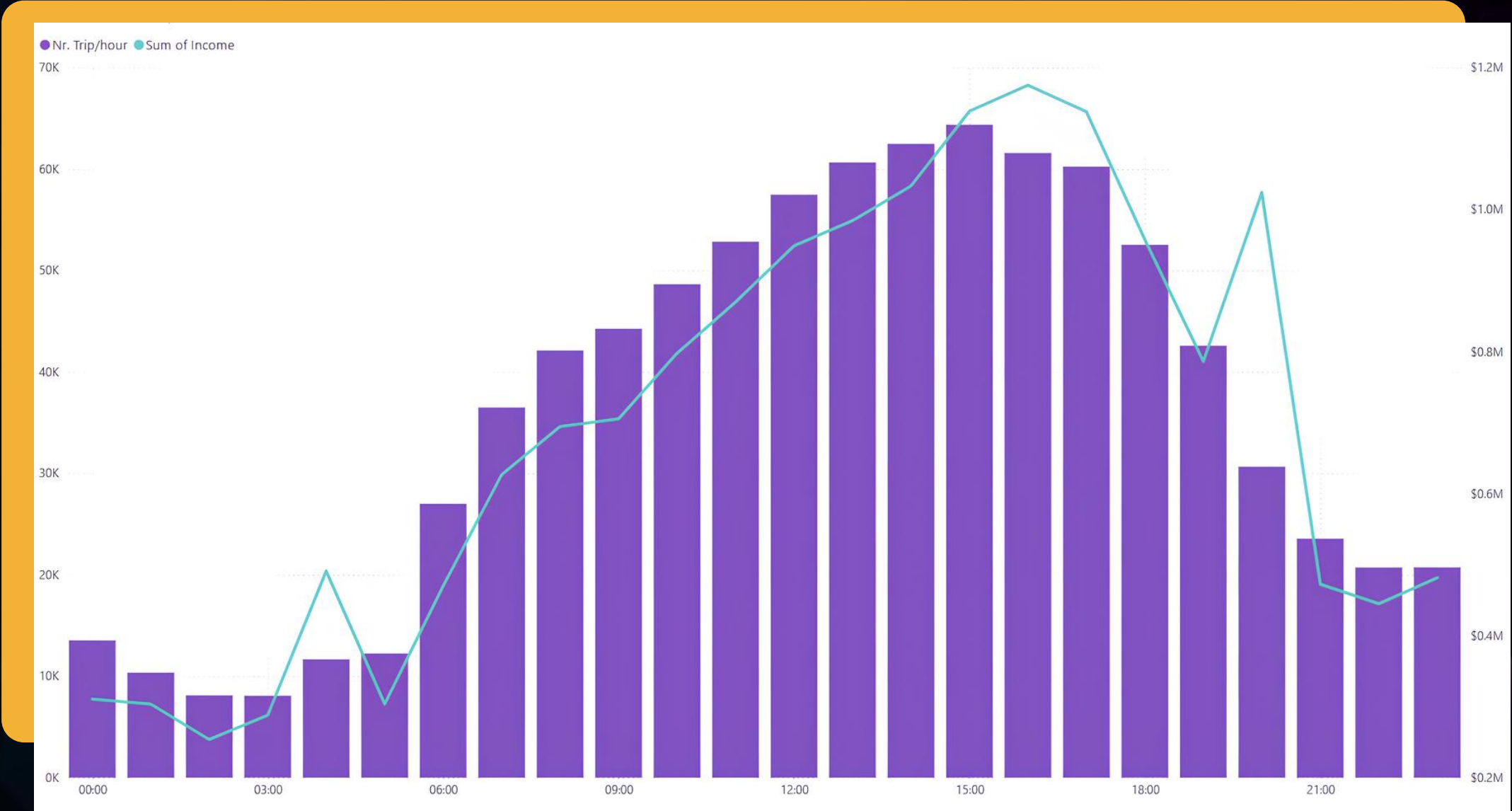




WHICH HOURS OF THE DAY
GENERATE THE MOST TRIPS AND REVENUE?

EMPLOYEE DASHBOARD

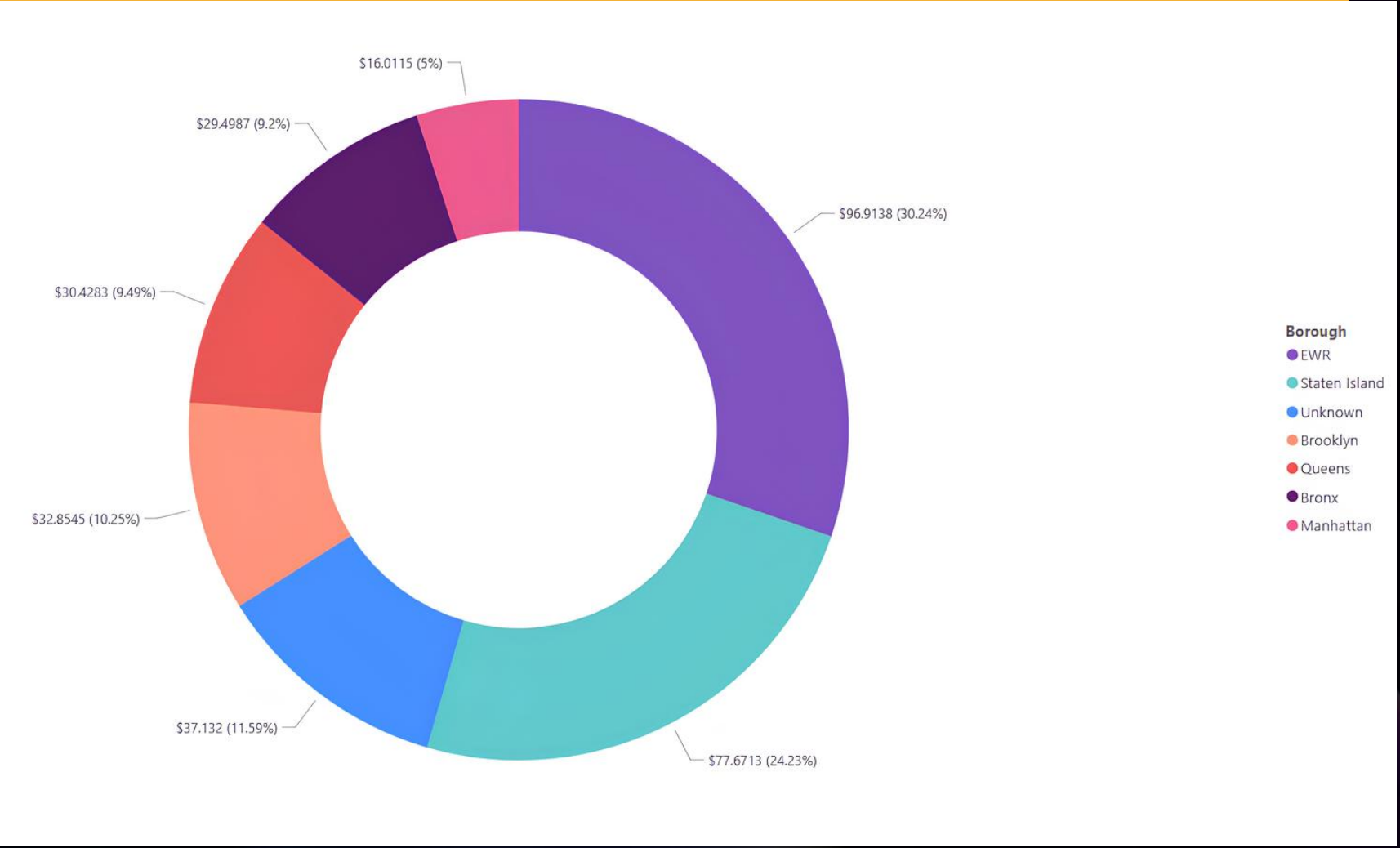
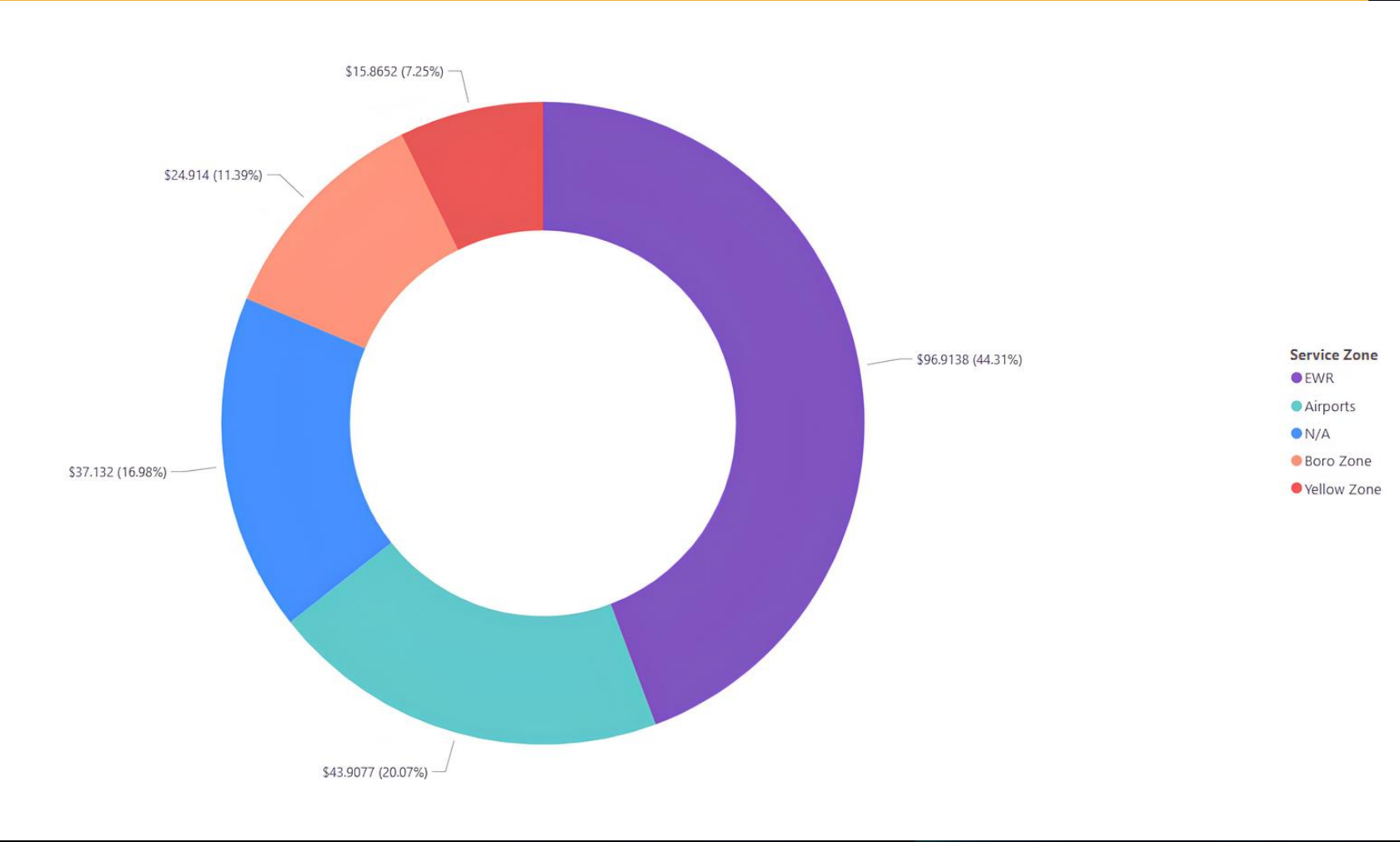
Focus effort on peak earning periods: 15:00–17:00 for highest trip volume and revenue, and 20:00 for fewer trips but higher-value fares, allowing similar earnings with less activity.



EMPLOYEE DASHBOARD



WHICH ZONES GENERATE THE HIGHEST AVERAGE INCOME PER TRIP?



EWR and Staten Island show the highest average income per trip, so prioritize pickups in these zones to maximize earnings.



PRINCIPAIS CONCLUSÕES

`tip_per_mile = DIVIDE('TRIPDATA MAI-JUN 2020'[tip_per_mile],
2020'[trip_distance], 0)`



ABORDAGEM ORIENTADA POR DADOS

O projeto destaca a importância de usar dados para melhorar a gestão dos serviços de transporte.

PADRÕES DE PROCURA E RENTABILIDADE

Identificaram-se variações claras na procura e rentabilidade conforme o tipo de veículo, zona e horário.

ALINHAMENTO ENTRE VENDOR E EMPLOYEE

Decisões integradas entre vendor e employee beneficiam todas as partes envolvidas no serviço.

USO EFICAZ DO POWER BI

O Power BI facilitou a transformação de dados complexos em informação clara e prática.



OTIMIZAÇÃO DE MOTORISTAS

Alocar motoristas estrategicamente nos horários de pico para maximizar a eficiência e atender à demanda elevada.

DISPONIBILIDADE DE VEÍCULOS MAIORES

Aumentar veículos maiores nos fins de semana para suportar maior volume de passageiros e melhorar o serviço.

FOCO EM ZONAS DE ALTA DEMANDA

Concentrar esforços operacionais em áreas com alta procura para otimizar recursos e rentabilidade.

DECISÕES BASEADAS EM DADOS

Utilizar análises de dados para melhorar a eficiência, experiência do cliente e maximizar lucros sustentáveis.

RECOMENDAÇÕES FINAIS





Q&A SESSÃO

Façam as vossas
perguntas





OBRIGADO!

APRESENTAÇÃO - 2026

